TAYLOR'S UNIVERSITY

Wisdom · Integrity · Excellence

SCHOOL OF COMPUTER SCIENCE

DEGREE PROGRAMMES

TEST (Final Piece) 30%

February 2025 SEMESTER

MODULE NAME	**: Cognitive Computing and Applications**
MODULE CODE	**: ITS69404**
DATE	**: 26th February 2025**
SUBMISSION DEADLINE	**: 26st February 2025, 10:00am – 8:00 pm, MALAYSIA TIME UTC+8**

Instruction to Candidates:

1. This cover page should be the first page of the Test answer report.
2. Give the answers for each question by just mentioning the relevant question number in the answer report. Do not reproduce the questions or the marking rubrics in the answer report.
3. Before submitting the final answers, ensure that the similarity index for this assessment **is less than 20% (overall) and 5% (single source).**

Student declaration:	
I certify that this assessment is entirely my own work, except where I have given fully documented references to the work of others, and that the material contained in this assessment has not previously been submitted for assessment in any other formal course of study.	
Name: Chew Ruo Ren	Student ID: 0369452
Signature: rr	

Question 1

a) Ethical dilemmas with autonomous drones in military operations include gray-morality in decision making, privacy issues, vulnerability to hacks and malfunctions, and future reliance on autonomous machines in security and law-enforcement. Autonomous drones only have directives and prohibitions in a warzone or covert operations. This can cause civilian casualties and accidents without human intervention should the drone perceive a civilian or hostages as a threat such as with wars where child soldiers are potential combatants. Authorities can potentially perform unauthorized and illegal surveillance on domestic and international civilians during wartimes for the country's supposed interest or personal gains within corrupted individuals. Machines are prone to cyberattacks and with military drones the dangers of a successful attack will lead to casualties and empowering criminal organizations or an opposing country at war.

Unmanned Aerial Vehicles (UAVs), a type of autonomous combat drone design in 2015, The Hacker News. https://thehackernews.com/2015/08/weaponized-drones-legal.html

As the world leans towards automating military operations and to a further extent, law enforcement, more machines will be built to automate security which will further complicate the existing ethical dilemmas and reduce job opportunities as well as giving future generations a binary imprint on morality due to being influenced by the enforcement of these machines. Countries and developers should regulate the abilities and the use of autonomous drones to

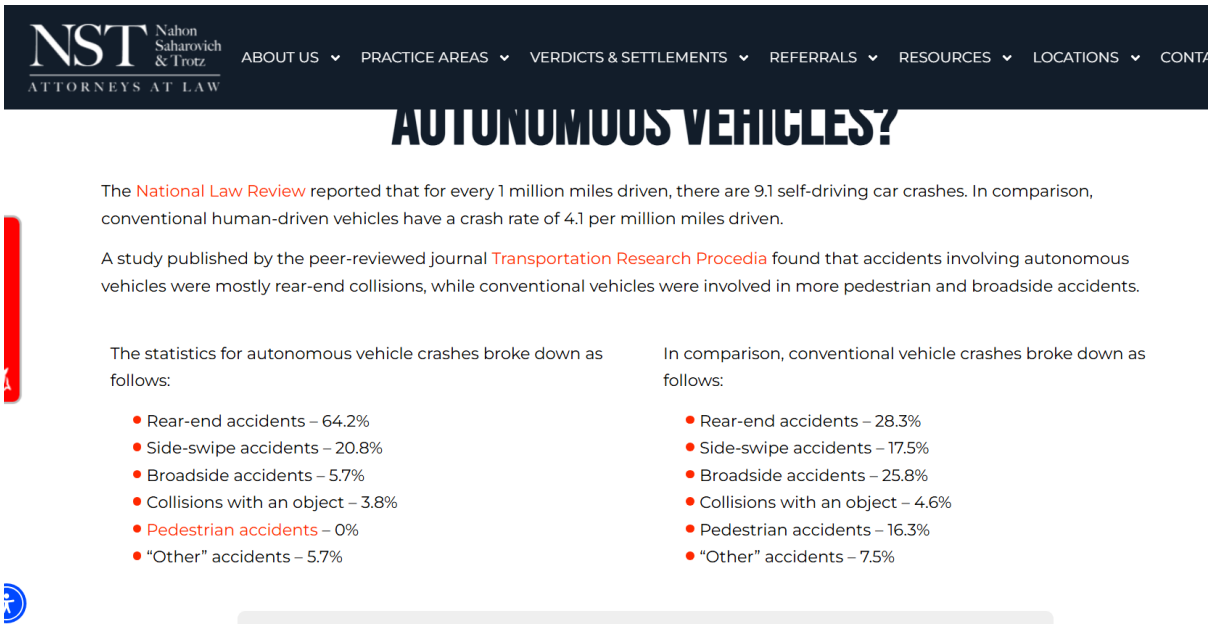
prevent unsanctioned casualties such as limiting the use and production of weaponized drones. The use of military drones should be entrusted through the authorization of multiple approvals so that these autonomous drones are reserved for last-resorts and reduce the potential damage done by autonomous drones. The world should consider drone uses as a part of the Geneva convention so that countries are less likely to depend on drones for basic law enforcement and regulate the occurrence of international surveillance.

b) Sensitive data such as customer's banking details and address are vulnerable to cyberattacks, drones may also be programmed to analyze consumer behavior and provide data to companies without legal consent of the consumer. Drones equipped with cameras can record consumers without consent or purpose and create new sensitive data to be accessed by other parties. Consumer behavior can be exploited by companies to target consumers with invasive advertising and unauthorized selling of data to third parties. Businesses will begin to exploit consumer personal data to generate profit and increase sales insight. Delivery drones can be equipped with cameras for navigation but can also record consumers without authorization. Recordings should be properly regulated and handled with by ensuring consumer privacy with temporary memory to ensure that the package has a confirm delivery whilst not recording or creating sensitive data to be stored.

c) Autonomous machines have no critical thinking to make emergency decisions during disaster responses as their decision-making abilities are binary. This may cause drones to prioritize individuals with lesser risks and abandon lives if deemed too troublesome for rescue. As the medical and emergency response industry becomes more privatized, drones will also prioritize clients and investors during an emergency response rather than saving lives unbiased. Medical services may attempt to save up on resources and provide minimal support to disasters due to reliance on autonomous machines as they reduce the dispatch of human operators.

Question 2

a) Manufacturers can make AI systems toggleable via user's input as a way of consent for the user and allows for human intervention to entrust autonomous services during driving. Autonomous driving will not be always fully in control with user consent as users can decide the appropriate timing to rely on autonomous driving such as traffic jams and slow neighborhood-area driving.



NST Nahon Saharovich & Trotz ATTORNEYS AT LAW

ABOUT US PRACTICE AREAS VERDICTS & SETTLEMENTS REFERRALS RESOURCES LOCATIONS CONTA

AUTONOMOUS VEHICLES?

The National Law Review reported that for every 1 million miles driven, there are 9.1 self-driving car crashes. In comparison, conventional human-driven vehicles have a crash rate of 4.1 per million miles driven.

A study published by the peer-reviewed journal Transportation Research Procedia found that accidents involving autonomous vehicles were mostly rear-end collisions, while conventional vehicles were involved in more pedestrian and broadside accidents.

The statistics for autonomous vehicle crashes broke down as follows:

- Rear-end accidents – 64.2%
- Side-swipe accidents – 20.8%
- Broadside accidents – 5.7%
- Collisions with an object – 3.8%
- Pedestrian accidents – 0%
- "Other" accidents – 5.7%

In comparison, conventional vehicle crashes broke down as follows:

- Rear-end accidents – 28.3%
- Side-swipe accidents – 17.5%
- Broadside accidents – 25.8%
- Collisions with an object – 4.6%
- Pedestrian accidents – 16.3%
- "Other" accidents – 7.5%

Statistic of autonomous and conventional vehicles crashes in 2021, NST Attorney at Law https://www.nstlaw.com/guides/self-driving-car-statistics/

With these statistics, manufacturers could also be more upfront with their test data and display them to consumers to provide trust and assurance of continuous improvement of the AI systems. This helps consumers to understand the risks and limitations of the AI system current capabilities and use autonomous driving more responsibly. AI systems should be able to contact emergency services autonomously in an event of an unavoidable accident. Manufacturers can employ a surveillance service for autonomous vehicles based on the vehicles operational data so that staff or AI system can identify crashes and contact emergency services at the earliest possible notice without using direct video surveillance on consumers and increase accident survivability.

b) Three privacy and security risks regarding personal and locational data are cyberattacks from unethical hackers, inside leaks within staff, and unconsented selling of personal data to other organizations. Data is always prone to hacks and require adequate cybersecurity services from

manufacturers to ensure consumer safety, which the manufacturers should invest in to improve their security such as monitoring tools and employ entrusted cybersecurity specialist. Manufacturers should perform thorough background checks on staff to minimize risk of inside leaks within the company so that consumers are not exploited and enforce customer data protection so that private data is not stolen. Manufacturers should also be responsible in handling the data and should be subjected to laws and regulations to prevent selling personal data to interested parties, and properly inform consumers of their intentions of managing their personal data.

c) Autonomous vehicles impact accessibility for underserved communities not just with the vehicle's affordability but also the accessibility of driving and its related laws. Autonomous vehicles will always be expensive to acquire due to high production costs. But autonomous driving will alter road conditions and etiquettes, requiring more consideration to maneuver and manage. Manufacturers can consider partnering with government programs and regulations, regulating prices according to the country's economy and incorporate with working benefits to not only provide motivation for employability but also provide a fair and affordable pricing of vehicles in lower GDP rate countries. These partnership opportunities can also enable governments to improve their infrastructure to accommodate for autonomous vehicles as autonomous vehicles depend on Internet of Things such as Wi-Fi and GPS services. Manufacturers and governments should also collaborate with incorporating autonomous vehicles to accommodate manual vehicles on the road on new road laws and regulations and effectively keep citizens well-informed of these new laws. This helps other drivers be more cautious and learn to adjust driving behavior in respect with autonomous vehicles to minimize accidents. Laws and regulations of autonomous vehicles should also enforce autonomous drivers to be responsible with their use of their vehicles.

Question 3

a) Four ethical principles developers can consider are to maximize the prevention of accidents, misuse and abuse of related cognitive applications, provide proper information to consumers of personal data management and acquirement, the responsibility of handling consumer's personal data, and understanding and discussing the limitations and potential dangers of the cognitive system. Developers should prioritize the security of their systems when suggesting new updates or during production to prevent any future threats of the system, flaws and potential errors can be used as a problem statement for development of new features and developers can easily identify problems with the current system for fixes. Developers should properly inform consumers or product testers about their personal data management and respect the user's consent and considerations, and only collect data needed for the system to minimize intrusion and respect the user's privacy. Developers should secure the data obtained from consumers by considering cybersecurity services and so that their personal lives are not at risk of attacks and leaks. Developers should actively discuss about the potential limitations of cognitive system and consider areas where human intervention is necessary so that developers can create dependable systems and improve the capabilities of cognitive systems.

b) Three privacy issues with cognitive systems are data sharing, data collection, and decision making. Developers should inform consumers about their data sharing scope if they have any and ensure that they share data to trusted and relevant parties that prioritizes the improvement of the cognitive system rather than purely based on profitability. Developers should also inform consumers about the scope of data collected and focus on relevant data only to not overstep boundaries and ensure that the consumer benefits from the system with the data collected during their use. Developers should also regulate the decision-making scope done by cognitive systems and avoid subjective topics for the cognitive system as cognitive system not capable of deciding the best-course of action, and should be recommended as suggestions or ideas for the consumers and also keep consumers inform of potential consequences of the decision made.

c) Three security measures developers can apply are data encryption, data access, and data backups. Developers should employ data encryption with their collected data as a basic security measure so that it is less susceptible to hacks, it can also serve as an additional security feature for the system to better appeal to consumers as well as increase privacy protection such as biometrics and protective protocols. Developers should ensure that data access is only available to relevant parties and employ multiple authentications to increase data security, this not only

protects the data from unauthorized access and sharing but as well as easy identification of intrusion under the event of attack. Developers can also backup relevant data so that data is not lost or displaced and allows the cognitive system to continue its operation using the backup data, providing a contingency if needed.